

ITAO303 - Mobile computing

CO4 - AT1 case study.

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16) Comparison of ios and android:-

Feature.	Ios.	Android.
Memory management	uses Automatic reference counting (ARC) for efficient memory management; but has stricter memory limits.	uses garbage collection through android Runtime, providing more flexibility for memory-intensive application.
Runtime environment	uses swift/objective C runtime with optimized native execution and strong hardware-software integration	uses Android Runtime with support for Kotlin and java, offering efficient application execution.
Process management	strictly controls background processes to save system resources and battery life.	Supports flexible background processing through foreground services and workmanager.

CPU utilization.	Highly optimized due to apple's integration of hardware and software.	Efficient CPU usage with support for multi-core processors and various hardware configurations.
Power management	Strong battery optimization with controlled background activities.	Uses Doze mode and app standby to reduce power consumption while allowing important background tasks.
Hardware support.	Limited to apple devices with consistent performance.	Supports a wide range of devices with different RAM and processor capacities.
Suitability for video editing.	Excellent performance but limited background processing.	Better support for high memory usage and long running video processing tasks.

Best choice :- Android .

- * Android is more suitable for a high - performance video - editing application .
- * It supports devices with large RAM capacities , which is useful for handling high - resolution videos .
- * Provides flexible background processing using foreground services and work manager
- * Supports powerful multi - core CPUs and GPUs for faster video rendering and encoding .
- * Gives developers greater control over CPU , memory , and background tasks .

Conclusion :-

- * Both iOS and android can support video editing applications .
- * iOS provides excellent performance

and strong Power management.

- * IOS has stricter restrictions on background Processing.

- * Android Provides more flexibility in memory, CPU and Process management.

- * Therefore, Android is the better choice for this application.